

Roots of Unity



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Abstract

The report discusses the solutions of $x^n - 1 = 0$. We show that if n and m have a common divisor then there are solutions to $x^n - 1 = 0$ that are also solutions for $x^m - 1 = 0$.

One reason for introducing complex numbers is that their introduction ensures that every n th order polynomial has n roots. For example, the polynomial equation:

$$x^n - 1 = 0$$

has n solutions, many of which are complex. Suppose that we construct a set containing all the solutions of the set of $x^n - 1 = 0$ equations that have $n \in \mathbb{N}$ and $n \leq N$. Given that each polynomial has n solutions it is not unreasonable to suppose that the cardinality of this set is given by equation 1.

$$N = \sum_{j=1}^N j = \frac{N(N+1)}{2} \quad (1)$$

However, as the black points in figure 1 show, when we do a numerical experiment using Sympy and construct a set containing all the solutions of the equations $x^n - 1 = 0$ with $n \leq N$ the cardinality of the set or roots grows more slowly with N than equation 1 (red points) predicts.

The reason the set of roots grows less quickly than equation 1 predicts is explained by 2. Many of the solutions of $x^n - 1 = 0$ are also solutions of $x^m - 1 = 0$ where $m \neq n$. For example, $x = 1$ is a solution to every single one of these equations. The cardinality of the set of solutions to the set of $x^n - 1 = 0$ equations with $n \leq N$ thus grows more slowly than equation 1 predicts because every element in a set needs to be distinguishable. You cannot construct a set containing N objects if two or more of those objects are indistinguishable.

To construct figure 2 we made a list of the roots of all the $x^n - 1 = 0$ equations with $n \leq 12$. Figure 2 shows how frequently each of the 46 solutions of these equations appear in this list. You can see that the most frequent solution is $x = 1$, which appears twelve times you as you would expect.

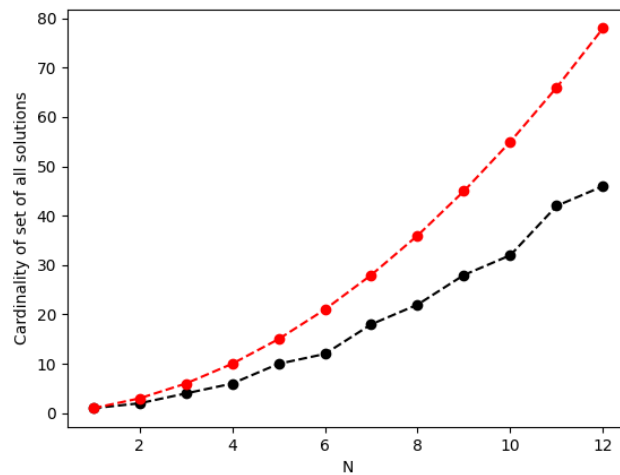


Fig. 1. The cardinality of a set containing all the solutions of all the equations $x^n - 1 = 0$ with $n \in \mathbb{N}$ and $n \leq N$. The black points shows the result of a numerical experiment that was performed using Sympy to determine how this cardinality grows as N increases. The red points show how the prediction of this growth that is obtained using equation 1. You can see that the actual growth in cardinality is slower than that predicted by equation 1.

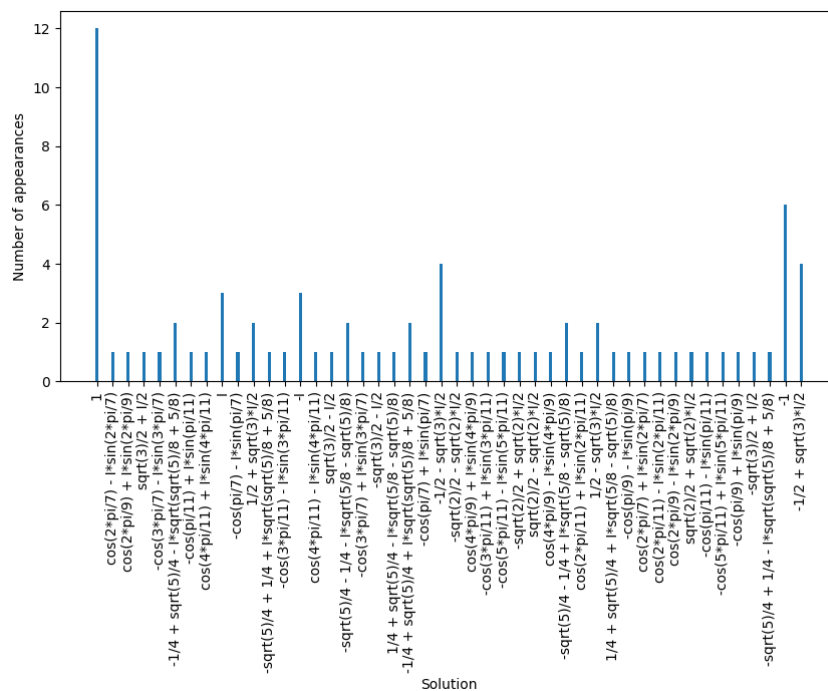


Fig. 2. Number of appearances of each of the 42 solutions of the set of $x^n - 1 = 0$ equations with $n \leq 12$.

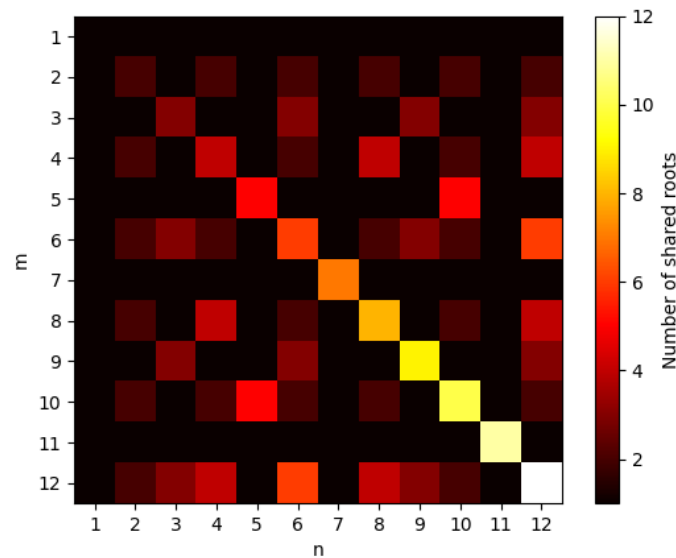


Fig. 3. Matrix indicating the number of solutions of $x^n - 1 = 0$ that are also solutions of $x^m - 1 = 0$. You can see that there is always at least one shared solution ($x = 1$) and that there is more than one shared solution if n and m have a common divisor.

An interesting experiment is to determine, how many times a solution of $x^m - 1 = 0$ is shared by $x^n - 1 = 0$. Figure 3 shows the matrix of results that is obtained from this analysis. The elements on the diagonal count the number of solutions that are shared when $m = n$. As the two equations are identical in this case, the solutions are identical and there are thus n shared solutions. The off-diagonal elements tell you how frequently a solution is shared if $m \neq n$. You can see that this quantity is always greater than or equal to one (as every pair of equations shares the solution $x = 1$). Furthermore, whenever m and n share a common divisor then the number of shared solutions is greater than 1.

The reason why $x^m - 1 = 0$ and $x^n - 1 = 0$ share solutions when m and n have a common divisor is obvious once you remember that the n solutions of $x^n - 1 = 0$ are uniformly distributed around the unit circle as shown in the top left panel of figure 4. For the case when $n = 12$ that is shown in the figure the 12 solutions sit at the vertices of a regular twelve-sided polygon. However, as shown in the remaining panels of this figure because 3, 4 and 6 all divide 12 we can also construct regular 3, 4 and 6 sided polygons from a subsets of the solutions of $x^n - 1 = 0$. The vertices of the hexagon are the solutions of $x^{12} - 1 = 0$ that are also solutions of $x^6 - 1 = 0$. The vertices of the square are the solutions of $x^{12} - 1 = 0$ that are also solutions of $x^4 - 1 = 0$. Lastly, the vertices of the triangle are the solutions of $x^{12} - 1 = 0$ that are also solutions of $x^3 - 1 = 0$.

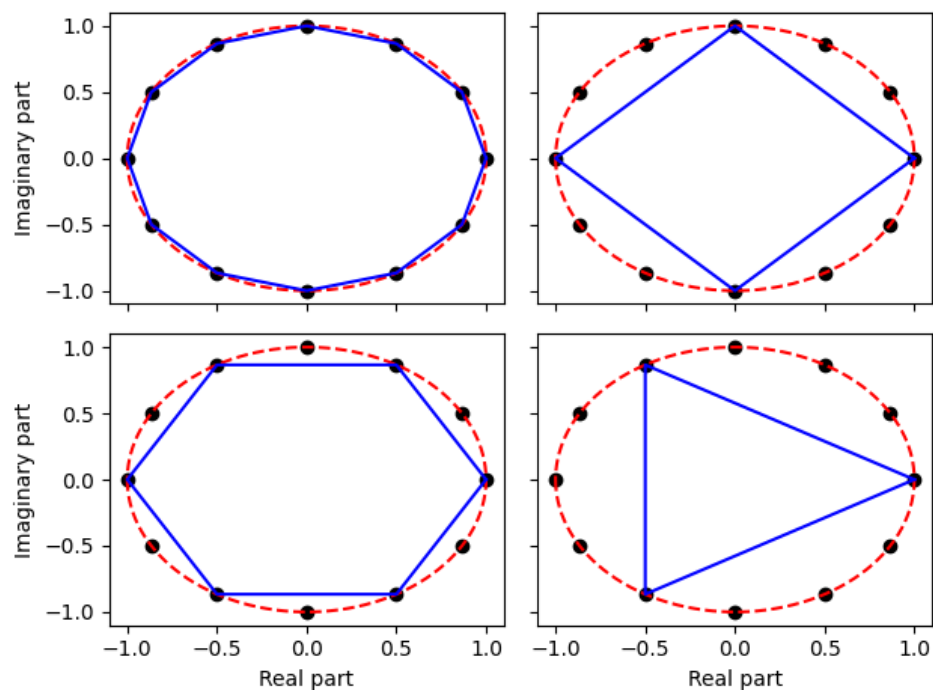


Fig. 4. The 12 solutions of $x^{12} - 1 = 0$ plotted in the Argand plane (black points). The blue hexagon in the lower left panel indicate the 6 solutions of $x^6 - 1 = 0$. The blue square in the upper right panel indicate the 4 solutions of $x^4 - 1 = 0$ and the blue triangle in the lower right show the 3 solutions of $x^3 - 1 = 0$. You can see that the solutions of these three equations are all also solutions of $x^{12} - 1 = 0$.